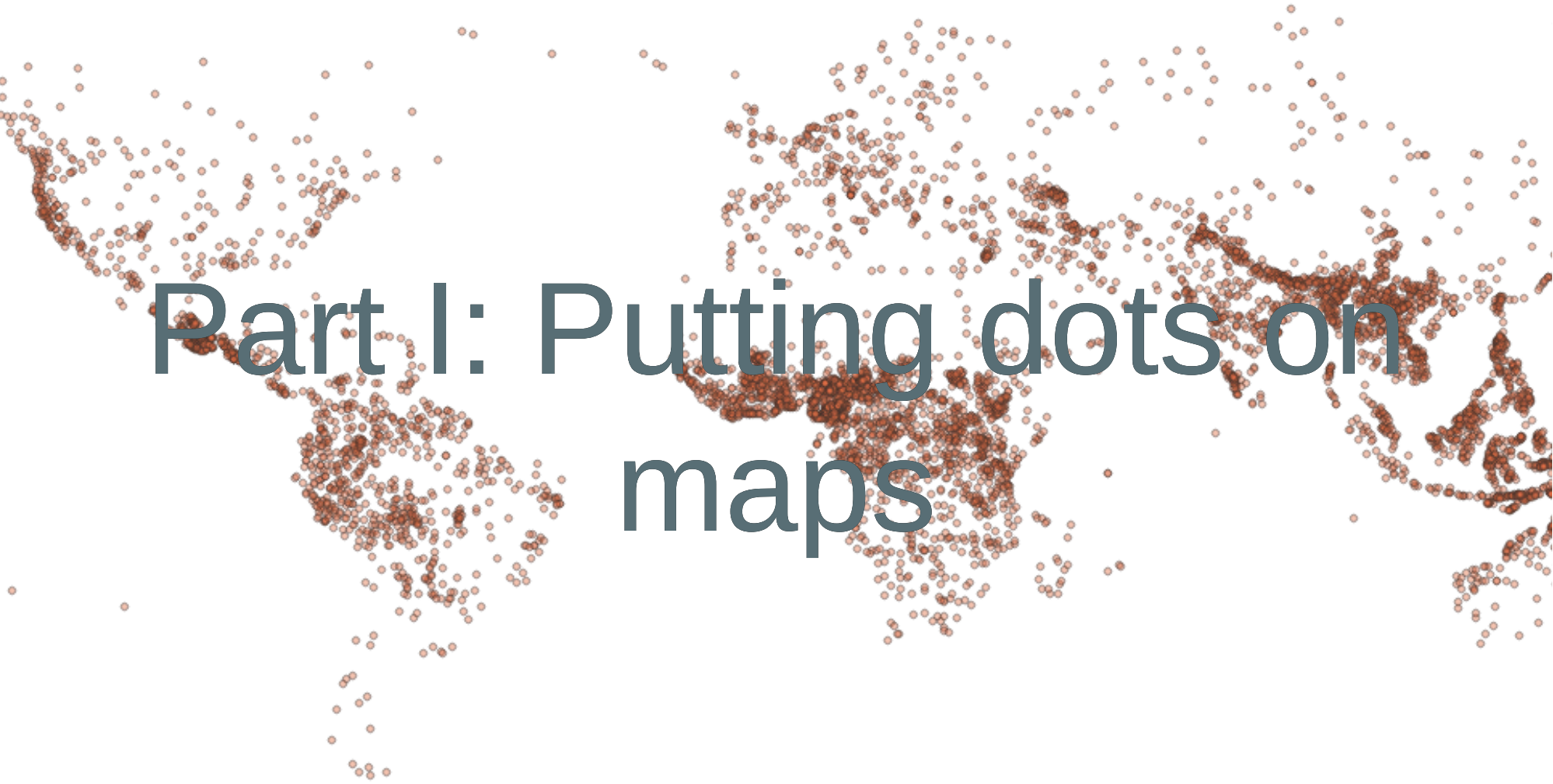
Connecting the dots

Growing an eco-system for cross-linguistic data

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Disclaimer

- I’m not a linguist - neither typological nor computational.
- This is my first keynote.



Part I: Putting dots on maps

Language Diversity

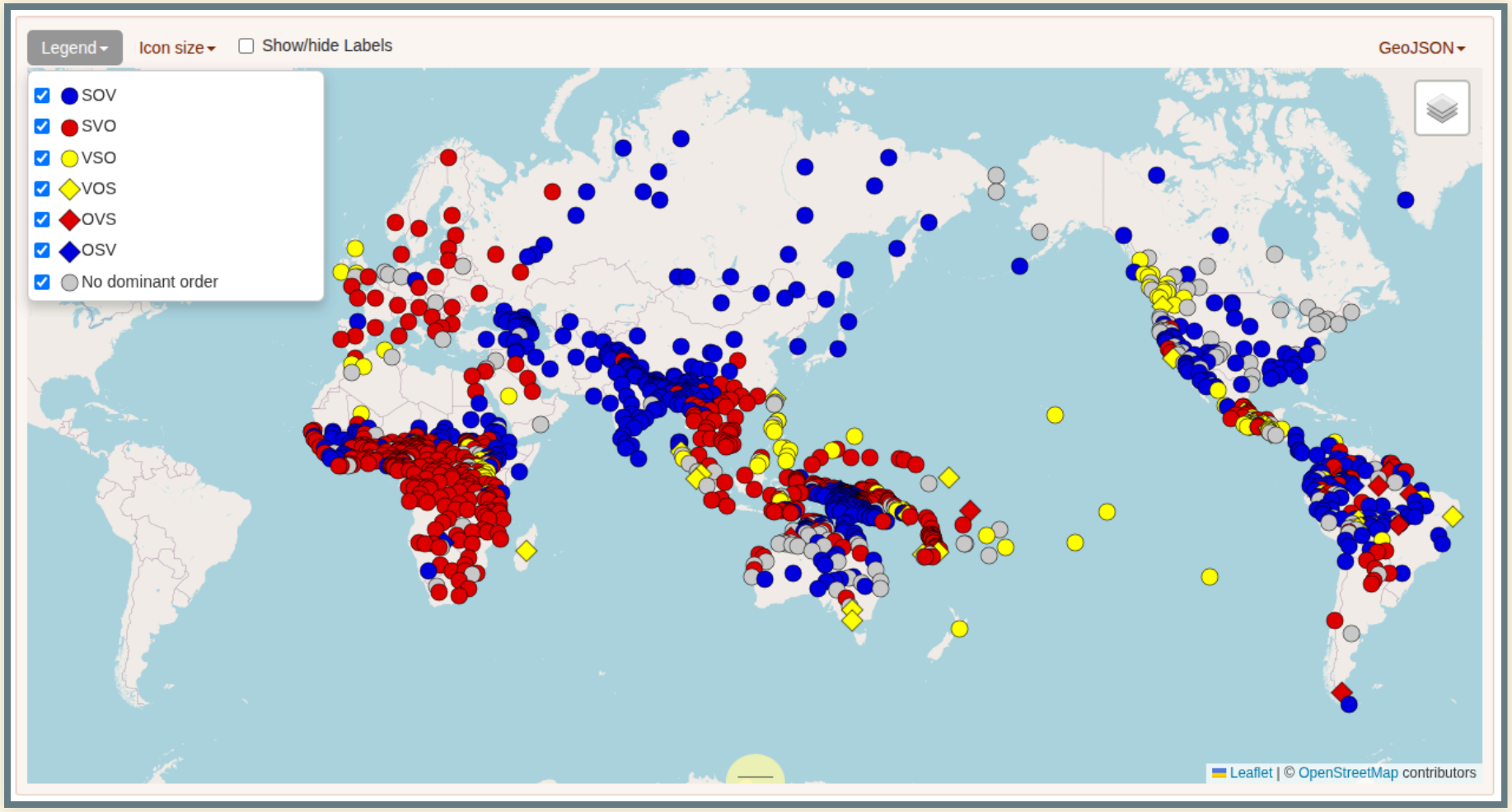
One of the most important aspects Typology can add to NLP is the diversity of the world's languages. As the sigtyp website puts it:

> *Linguistic Typology is the study of the diversity of the myriad of languages of the world as it is manifested in their grammatical, lexical, sociolinguistic or other properties.*

Diversity Linguistics

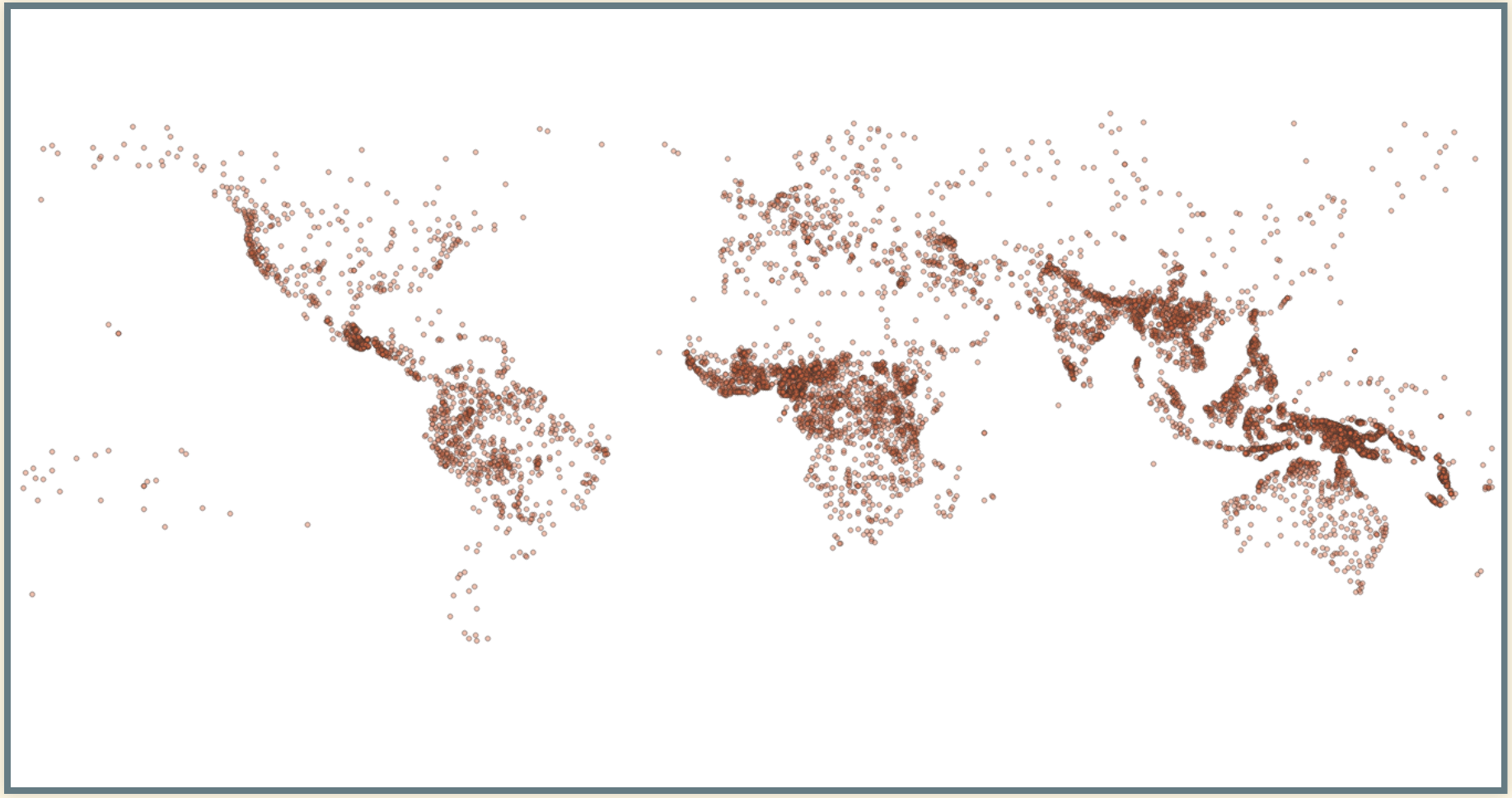
- term coined by researchers in Bernard Comrie's Linguistics department at MPI EVA
- not 10s or 100s of languages, but 1000s
- lots of Typology, but also Historical Linguistics, Sociolinguistics, etc.
- flagship project: WALS - the World Atlas of Language Structures.

The World Atlas of Language Structures



Everyone’s a collector

- Data collection is a hobby shared by many Linguists
- Data re-use is often “manual”, though
- The first need for computational re-use often only arises when collections get too big



Beyond visualization

- “Can I contribute my data to WALS?”
  → need to merge datasets!
- “Can I visualize my data like in WALS?”
  → need to use software with different datasets!
- “APiCS is just like WALS - except, it allows multiple feature values assigned to the same language”
  → need to figure out shared data model!

Interoperability - the Microsoft model

- If data is browsed through the web anyway, make the web application the “interface” to the data
- If everyone uses `clld` to publish their data, data would be interoperable (or at least could be accessed in similar ways)
- The data model can then be regarded as “implementation detail”

Web apps as data publications do not scale

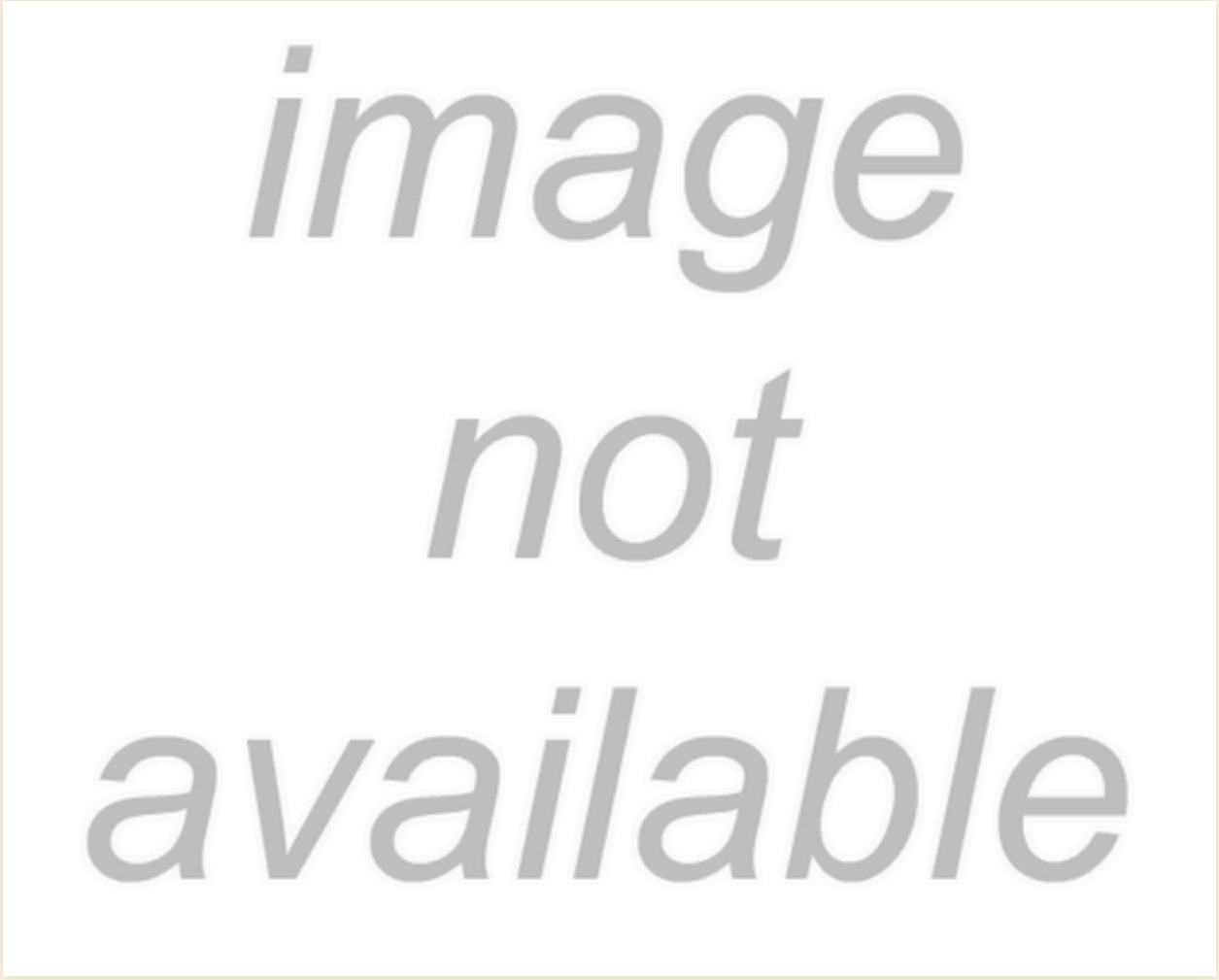
- Keeping web apps running does not require much, but it does require continuous support.
- Academic institutions are bad at providing long-term support.
- “Cool URLs don’t change” - cf. the average university home page.



Part II: Connecting the dots

The landing page for the “Atlas of the World’s Languages” on Routledge’s site features a quote attributed to Mark Pagel:

> *Routledge’s monolithic new Atlas of the World’s Languages provides an unprecedented account of the linguistic composition of the world… So large a contribution is the atlas that I predict it will stimulate its own cottage industry of geographical linguistic research.*


image
not
available

nature > scientific data > articles > article

Article | Open access | Published: 16 October 2018

Cross-Linguistic Data Formats, advancing data sharing and re-use in comparative linguistics

Robert Forkel ✉, Johann-Mattis List ✉, Simon J. Greenhill, Christoph Rzymski, Sebastian Bank, Michael Cysouw, Harald Hammarström, Martin Haspelmath, Gereon A. Kaiping & Russell D. Gray

Scientific Data **5**, Article number: 180205 (2018) | Cite this article

CLDF - cutting out the middleman

- a package format bundling tabular data in CSV files with metadata in a JSON file
- built on W3C's CSVW standard
- adding domain-specific semantics via an ontology of terms that can be referenced in a dataset's metadata.

CLDF - design decisions

- A CLDF dataset is just a bunch of text files, perfectly suited for version control with tools such as git.
- Multi-table datasets look somewhat complicated, but see Wilson et al. and Wickham 2014.
- It's based on `clld`'s internal data model, hence on years of experience with different datasets.

Wilson G, Bryan J, Cranston K, Kitzes J, Nederbragt L, Teal TK (2017) Good enough practices in scientific computing. PLoS Comput Biol 13(6): e1005510. DOI:10.1371/journal.pcbi.1005510

Wickham, H. . (2014). Tidy Data. Journal of Statistical Software, 59(10), 1–23. DOI:10.18637/jss.v059.i10

CLDF - core objects

- Tables (*components*): Language, Parameter (aka Feature), Value, IGT Example
- Macro-types (*modules*): Wordlist, StructureDataset, Dictionary, TextCorpus

“Simple things should be simple …”

- Exploratory data analysis with UNIX shell tools, e.g. `csvkit`

```
$ csvstat -c Glottocode wals/cldf/languages.csv
  6. "Glottocode"

        Type of data:          Text
        Contains null values:  True (excluded from calculations)
        Non-null values:       2645
        Unique values:         2503
        Longest value:         8 characters
        Most common values:    None (928x)
                               basq1248 (11x)
                               swis1247 (6x)
                               stan1295 (6x)
Row count: 3573
```

```
$ csvjoin -c Glottocode wals/cldf/languages.csv \
grambank-cldf/cldf/languages.csv | csvstat -c Glottocode
  6. "Glottocode"

        Type of data:          Text
        Contains null values:  False
        Non-null values:       1518
        Unique values:         1442
        Longest value:         8 characters
        Most common values:    basq1248 (11x)
                               tibe1272 (6x)
                               nort3139 (4x)
                               roma1326 (4x)
Row count: 1518
```

Detour: Language identification

For cross-linguistic data, language identification is a non-trivial problem!

- The *Atlas of the World's Languages* was published in 2007
- It references languages by name (and coarse classification)
- In 2025, some of the names used didn't lead to **any** hits on Google!

Old problem - not so old solutions

- Why didn’t the *Atlas* use ISO 639-3?
- And why did WALS mint three-letter language codes when there’s ISO 639-3?
- ISO 639-3 is from 2007, and Ethnologue’s “SIL codes” are from 1984 (and some changed before becoming “ISO codes”). That’s why also LinguistList and WALS minted three-letter codes - there was no standard there yet.

Why Glottolog?

- Classification, alternatives names and some coverage of dialects are essential for “retro-digitization” efforts
- The classification is the equivalent to GBIF’s *backbone taxonomy* for language diversity - a tool for data integration
- so, no reason to fight over classification details, but embrace it as essential tool!

Reference catalogs

Make complex data manageable by factoring out external data:

- Glottolog
- Concepticon
- CLTS
- all reference catalogs are available as versioned CLDF datasets!

“… complex things should be possible”: The eco-system

- CLDF specifies an interface between data and tools
- So data collection and work on methods can proceed in separation (which is important in a field where data collectors and data analysts are often different people)

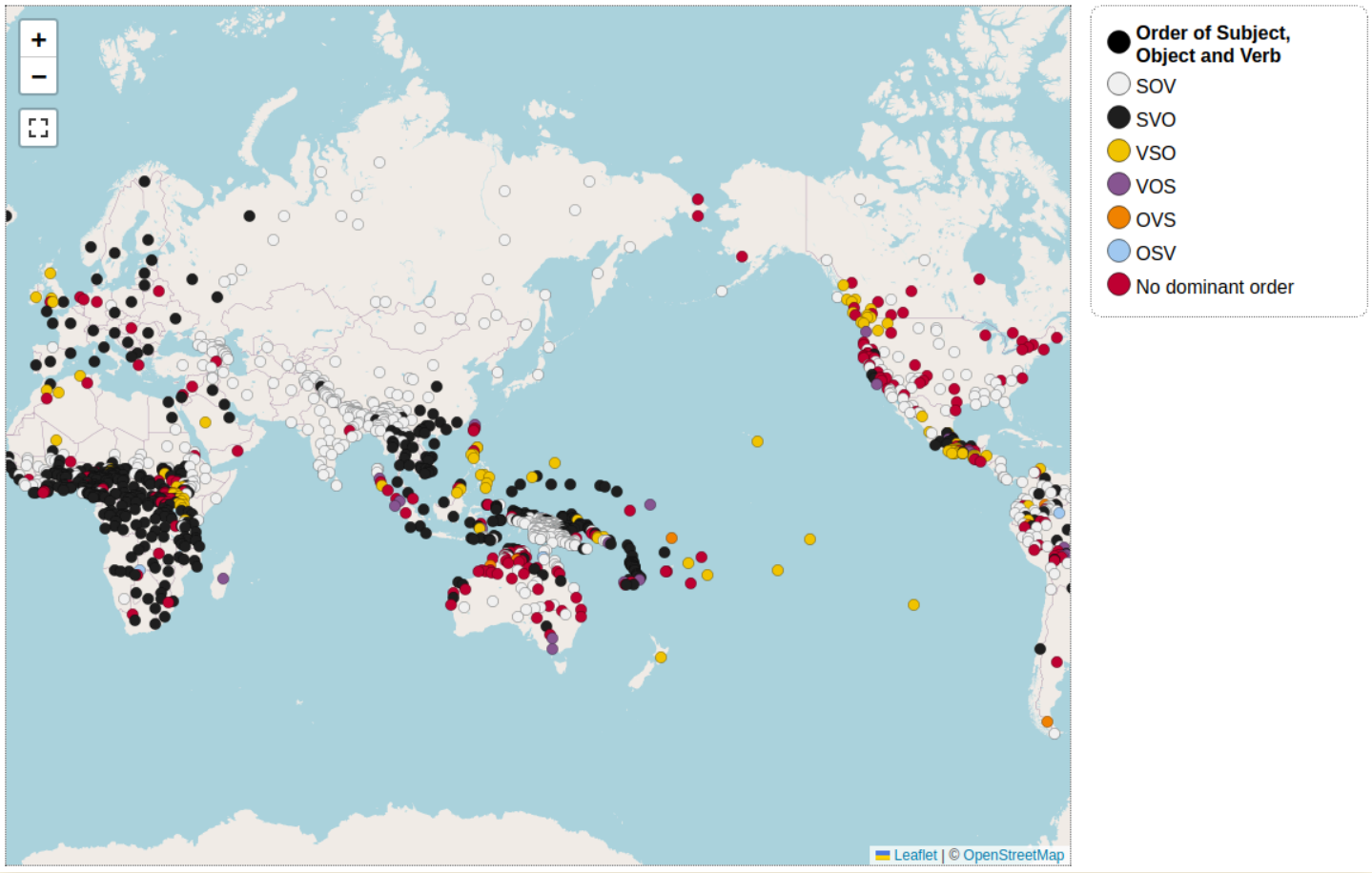
Tools

Implement complex things once and apply to all datasets:

- `pycldf`: read and write (and **validate**) CLDF data
- `cldfviz`: visualize CLDF data
- `cldfzenodo`: access published CLDF data

`cldfviz`: “Putting dots on maps” - without a server!

```
$ cldfbench cldfviz.map \
--parameter 81A --pacific-centered wals/cldf
```


+
−
Order of Subject, Object and Verb
SOV
SVO
VSO
VOS
OVS
OSV
No dominant order
Leaflet | © OpenStreetMap



Part III: A model to build on

`cldf createdb`

CLDF’s killer feature: Each CLDF dataset can be turned into an SQLite database!

The CLDF metadata provides all necessary information to create suitable relations between tables.

nature > nature human behaviour > articles > article

Article | Open access | Published: 24 September 2024

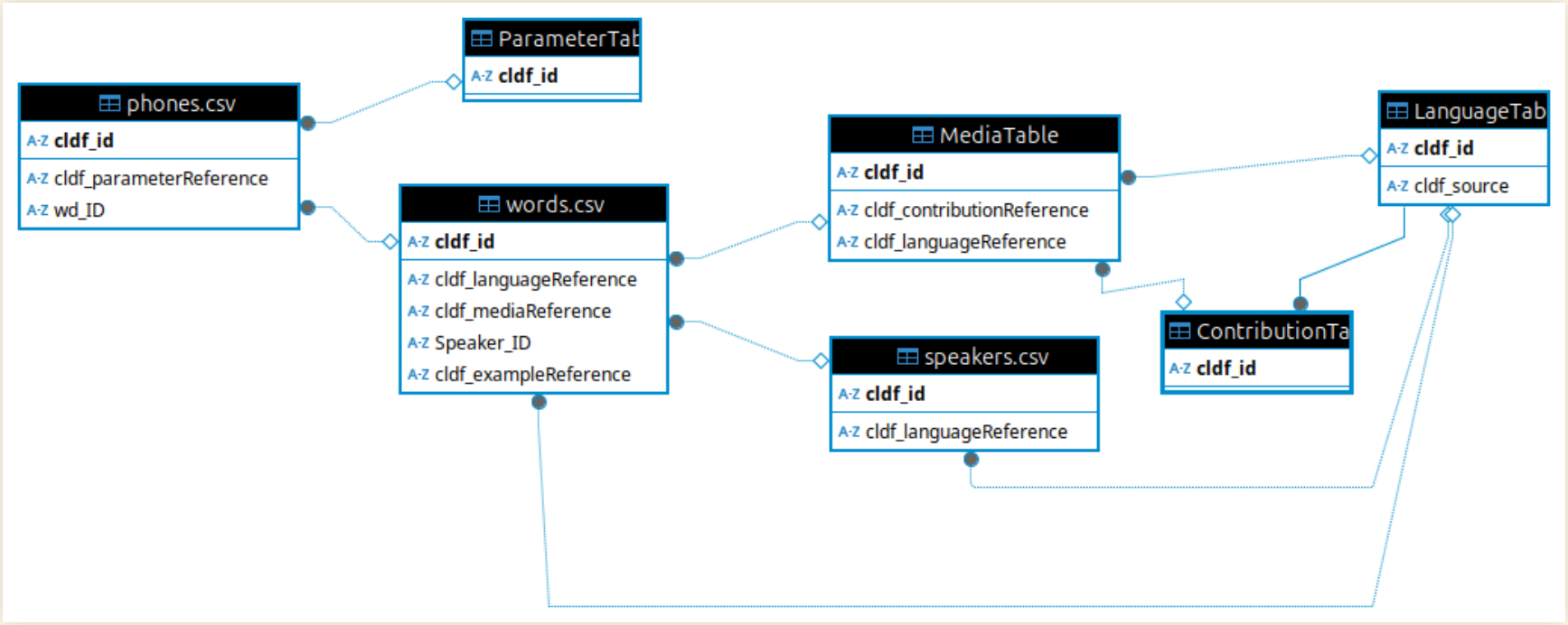
Consonant lengthening marks the beginning of words across a diverse sample of languages

Frederic Blum ✉, Ludger Paschen, Robert Forkel, Susanne Fuchs & Frank Seifart

Nature Human Behaviour **8**, 2127–2138 (2024) | Cite this article

Blum et al. 2024

- no spec for time-aligned text in CLDF yet
- but CLDF can still be a good fit:
- the DoReCo corpora are available as CLDF dataset
- using an ad-hoc data model for the phones
- SQLite allowed us to handle ~2mio time-aligned phones, including metadata joined from a couple of other tables with ease!


ParameterTab
cldf_id
phones.csv
cldf_id
cldf_parameterReference
wd_ID
words.csv
cldf_id
cldf_languageReference
cldf_mediaReference
Speaker_ID
cldf_exampleReference
MediaTable
cldf_id
cldf_contributionReference
cldf_languageReference
LanguageTab
cldf_id
cldf_source
ContributionTa
cldf_id
speakers.csv
cldf_id
cldf_languageReference

“Language polygons”

`http://cldf.clld.org/v1.0/terms.rdf#speak`

A mini-spec to make “language polygons” in CLDF dataset easily accessible.

- another way of “connecting the dots”:
- `cldfgeojson`: cross-validate polygons via Glottolog data

nature › scientific data › data descriptors › article

Data Descriptor | Open access | Published: 29 August 2024

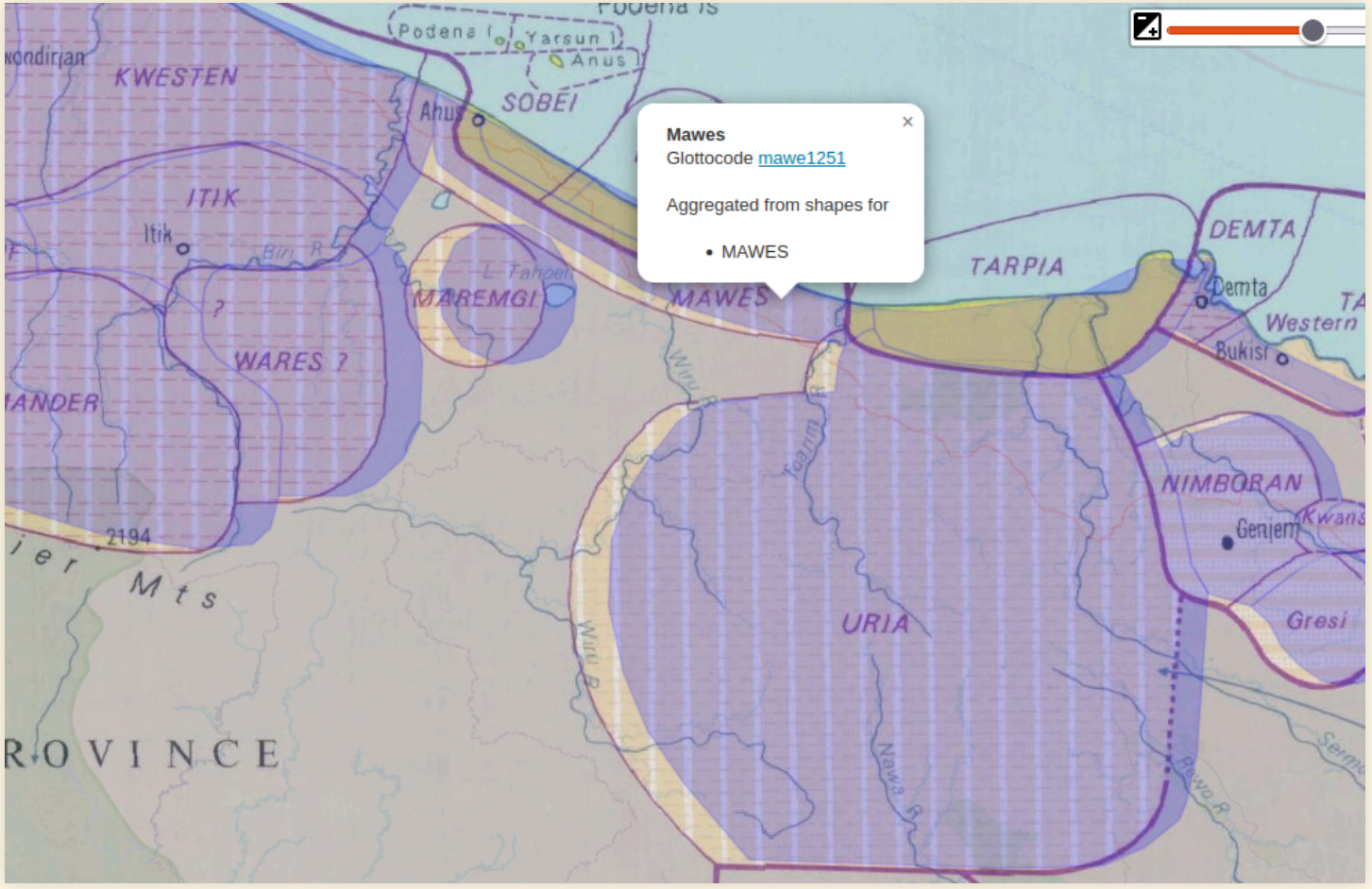
A revised digital edition of Wurm & Hattori's Language Atlas of the Pacific Area

Robert Forkel ✉ & Harald Hammarström

Scientific Data **11**, Article number: 949 (2024) | Cite this article

Forkel and Hammarström 2024

- Polygons are often a better proxy for “where a language is spoken” than point coordinates
- e.g. for computing probability of borrowing
- or correlations with environmental factors


Podena I
Yarsun I
Anus
SOBEI
KWESTEN
ITIK
Itik
Biri R
MAREMGI
L Tahoen
MAWES
TARPIA
DEMTA
Demta
Western
Bukisi
WARES ?
NIMBORAN
Genjem
Kwans
URIA
Gresi
2194
Mts
ROVINCE
Mawes
Glottocode mawe1251
Aggregated from shapes for
MAWES

Comparative Dictionaries

aka “Etymological Dictionaries”

- “Cross-concept” cognate sets
- Data about semantic change
- Data about sound change
- ACD (Austronesian Comparative Dictionary), PMED (Preliminary Mayan Etymological Dictionary), CSD (Comparative Siouan Dictionary)

nature > scientific data > data descriptors > article

Data Descriptor | Open access | Published: 17 June 2025

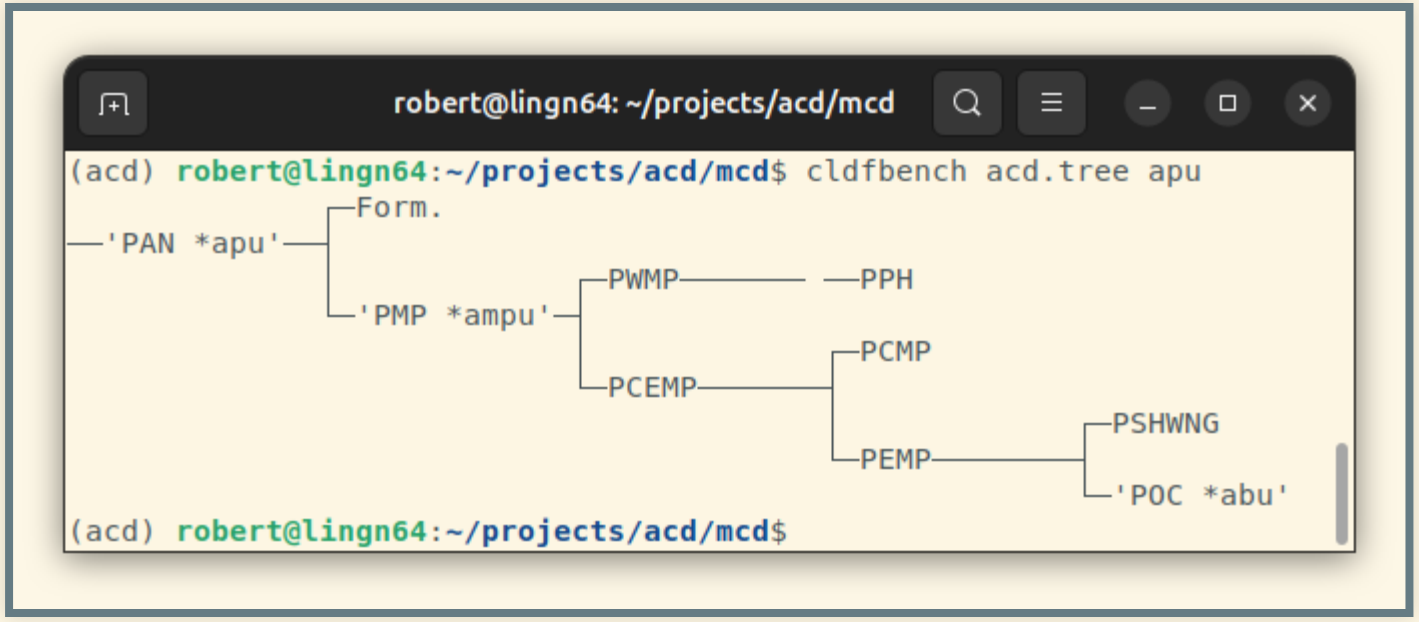
The Austronesian and the Micronesian Comparative Dictionaries as CLDF datasets

Alexander D. Smith ✉, Robert Forkel ✉ & Lev Blumenfeld

Scientific Data **12**, Article number: 1015 (2025) | Cite this article

Smith, Forkel and Blumenfeld 2025

- CLDF already has support for language phylogenies
- Reconstruction of protoforms assumes a tree of protolanguages
- We bundle the tree with the cognate sets from comparative dictionaries
- Thus, providing a dataset with lots of instances of form changes with known directions


robert@lingn64: ~/projects/acd/mcd
(acd) robert@lingn64:~/projects/acd/mcd$ cldfbench acd.tree apu
Form.
'PAN *apu'
PWMP
PPH
'PMP *ampu'
PCMP
PCEMP
PSHWNG
PEMP
'POC *abu'
(acd) robert@lingn64:~/projects/acd/mcd$

The datasets

- The reference catalogs: Glottolog, Concepticon, CLTS
- Typological databases: WALS, Grambank, etc.
- Multilingual wordlists: lexibank, ASJP
- Interlinear Glossed Text: IMTVault
- Annotated corpora: DoReCo + MultiCAST
- language phylogenies from phlorest
- “language polygons” from glottography

The datasets

A (still incomplete) list of (~300) Zenodo uploads containing CLDF datasets is avilable at https://meta.clld.org/zenodoconcepts

Take home messages

There’s plenty of cross-linguistic data out there, to quote van Esch et al. again

> *it’s not that there aren’t any resources, it’s just that they need to be unlocked if there is a desire to build technology*

and CLDF is a tool to unlock these resources.